

Linux Programming: Assignment-6:

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SECTION- C

1. Which command is used to list the contents of a directory?

Justify with proper example.

Ans:

Command: ls

ls command stand for list

It displays all the files and sub directorues

It provieds detailed information making it very versitalie for inspecting files in linux

2. Write the command to create a new directory named 123test_dir.

Command: mkdir

We can create the 123test_dir by command mkdir 123test_dir

It stands for make directory

Ccreates a folder with specified name

3. What is the purpose of the sed command? Justify with proper example.

Ans:

The sed command is used for stream editing where we can find,replace,insert,or delete text in files

Purpose of sed command

- Text substitution/replacement
- Deleting or inserting lines
- Performing batch edits on files
- Filtering and transforming text in pipelines

Example:

Hello World

Hello Linux

command :sed 's/Hello/Hi/' file.txt

output :

Hi World

Hi Linux

Instead of manually editing we can use sed command to transform a complex editing in one single command

4. Which distinct command is used to display one-line descriptions of any commands?

Ans:

Command: whatis

It provides a small detailed description of the specified command it is ideal for quick reference

5. Write the command to create an empty file named “notes.txt”.

Ans:

Command: touch notes.txt

This command creates empty files, if the file does not exist it will be created as an empty files

6. Differentiate between grep and awk commands with an example.

Ans:

grep command is used to search a specific patterns in files or input it filters lines containing the pattern

Awk command is a text processing tool used to search,extract,manipulate and format data from text files

Example for grep command

apple 10

banana 20

cherry 15

command : grep "banana" file.txt

Output:

banana 20

command: awk '\$2 > 12 {print \$2}' file.txt

Output:

20

15

7. Write the command to give read, write, and execute permission to the owner of a file script.sh.

Ans:

Command:

chmod u+rx script.sh

Example:

```
chmod u+rwx script.sh  
ls -l script.sh
```

Output:

```
-rwxr--r-- 1 user user 1024 Oct 11 22:00 script.sh
```

8. How is chown different from chgrp? Give one example for each.

Ans:

Chown command:

This command changes the owner of a file or directory

Example:

```
chown note file.txt
```

```
check:ls -l file.txt
```

```
Output:-rw-r--r-- 1 alice users 1024 Oct 11 22:00 file.txt
```

chgrp command:

This command changes the group ownership of a file or directory

Example:

```
chgrp notes file.txt
```

```
check:ls -l file.txt
```

```
output:-rw-r--r-- 1 alice developers 1024 Oct 11 22:00 file.txt
```

9. A user complains that they cannot execute a file even though it exists in their directory. How would you troubleshoot this using ls -l, chmod, and whoami?

Ans:

1. Check the file permission by the ls -l command

```
ls -l filename
```

2. Check the current user, use the whoami command

Whoami

3. Add execute permission if needed by using the chmod command to give execute permission

4.verify it again by the ls -l command after adding the execute

10. Design a command pipeline to: find all .log files modified in the last 2 days in /var/log, display them on screen, and save the results into a file recent_logs.txt using tee command.

Ans:

Command:

```
find /var/log -name "*.log" -mtime -2 | tee recent_logs.txt
```